Ray-Junior Kgomotso Mongwe

Software Engineer

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Professional Summary

Software Engineer at Afrosurance, an early-stage fintech startup, one of the first developers on the team and the sole owner of testing and quality assurance. Chose and implemented the testing stack (Playwright and Jest) across both new and legacy systems, while building across the full stack with Vue, React, Next.js, Node.js and MongoDB. Works directly with the co-founder/CTO and enjoys wearing many hats: shipping polished, user-centric products, writing clean code, and making sure what ships is reliable.

Technical Skills

- **Testing & QA:** Playwright, Jest, Cypress, Puppeteer, Burp Suite
- **Languages & Core:** JavaScript, TypeScript, Python, Java, C, C++, C#, HTML5, CSS3, SQL
- **Frontend & Frameworks:** Vue.js, Vuetify, React, Redux, Next.js, Nuxt, Tailwind CSS, Bootstrap, Sass
- **Backend & APIs:** Node.js, Express.js, Django, Flask, ASP.NET, GraphQL, Apollo Server
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB, Mongoose
- **Cloud & DevOps:** Azure, AWS, Docker, Kubernetes, Vercel, Railway, Render, Git, Bash

Professional Experience

Software Engineer | Afrosurance | September 2025 – Present

- One of the early engineers (not a founding member), working hand in hand with the co-founder/CTO to design, build and maintain enterprise software and backend services.
- Sole owner of all testing and quality assurance: selected the testing stack (Playwright and Jest) and implemented it on new systems as well as on older systems built before I joined.

- Build and optimise internal tools and user-facing applications that streamline operational workflows.
- Contribute across frontend, backend and testing in a startup environment, taking on whatever the product needs next.
- Started with the company as a freelance developer in June 2025, then joined full time as a Software Engineer in September 2025.

Featured Projects

Afrosurance Platform | Vue 3, Vite, Vuetify, Axios, JavaScript

- Sole frontend developer on a fintech platform bridging the gap between formal financial services and the African rural and informal economy. The backend (Node.js, MongoDB, GraphQL) was built by the wider team.
- Built the entire frontend: components, integration with the REST API, and the dashboard UI, with custom Vuetify branding, routing, and deployment on Vercel.
- Drove product direction by proposing a phone/device insurance option based on the needs of the audience the platform targets.

Afrosurance Client Test | Vue 3, Vite, Tailwind CSS, Axios, Node.js, Express, MongoDB, Mongoose

- Designed and built a full-stack online assessment platform end to end: Vue 3 frontend, Express REST API, MongoDB data layer, and JWT-based admin authentication.
- Admins create timed tests, schedule when they open and close, and share them through access codes, links and QR codes; candidates can take a test on any device, with progress autosaved so a refresh never loses their place.
- Scoring runs on the server; admins get an analytics dashboard, full review history for every submission, batch access codes for tracking groups of candidates, and a CSV export hardened against spreadsheet formula injection.

Heart of Mopani | Next.js, React, Tailwind CSS

- Built a regional tourism platform highlighting the culture, nature and practical travel logistics of the Mopani district in Limpopo.
- Created reusable listing pages for places to stay, eat and things to do, plus featured experiences, a getting-there guide, and a "Pack for the Wild" gear guide with affiliate links.

- Used Next.js for server-side rendering to support SEO on a content-heavy site; responsive, mobile-first UI with Tailwind CSS.

Education & Certifications

- **Introduction to Computer Science (CS50x)** | HarvardX: algorithmic thinking, low-level memory management, data structures, and web development.
- **CS50 SQL** | HarvardX: designed and built complex relational databases, culminating in the Minecraft World Logistics Database project.
- **CS50 Cybersecurity** | HarvardX: core security principles, vulnerability assessment, and threat mitigation.
- **Fundamentals of AI** | Cognitive Class: core AI concepts, algorithms, and real-world machine learning applications.
- **DevOps with Docker** | University of Helsinki: containers, Docker Compose, and containerised development and deployment workflows.
- **DevOps with Kubernetes** | University of Helsinki: container orchestration, deploying and managing applications on Kubernetes clusters.
- **Full Stack Open** (currently enrolled) | University of Helsinki: modern web development with React, Node.js, GraphQL, TypeScript and testing.
- **CS50 AI** (currently enrolled) | HarvardX: search, machine learning, neural networks and natural language processing in Python.